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Implicit Inflation and Risk Premiums in the Brazilian Fixed Income Market

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ABSTRACT: The breakeven inflation, the differential between nominal and real yields of bonds, is often used as a predictor of future inflation. The model presented here decomposes this interest rate differential into a risk premium and implicit inflation using a parametric formulation based on no-arbitrage conditions using nominal and indexed yield curves in Brazil, via an affine model of the Nelson–Siegel family. The measures of implicit inflation obtained from the model are shown to be unbiased estimators of future inflation for short horizons and carry some information for long horizons, and the model forecasts are superior to market surveys.

KEY WORDS: affine term structure models, bond markets, inflation, risk premium

For the proper management of monetary policy, central banks should be interested in the inflation expectations of consumers, entrepreneurs, and financial market agents. These expectations are important in determining future inflation and in asset pricing procedures. In general, measures of future inflation are obtained from two main sources: expectation surveys and the information contained in financial market instruments. One important limitation is that data from market surveys are obtained at lower frequencies (monthly, quarterly, and annual forecasts), in contrast to the daily and intra-daily high frequency trading of financial assets. Another limitation of surveys is that usually they only provide market expectations for time horizons up to 1 year, and thus do not give information about long-term inflation expectations.

Due to the problems with expectations taken from surveys, different ways of extracting expectations implicit in financial assets have been proposed. A key work in this literature is Svensson and Soderling (1997), which discusses how to extract expectations contained in bonds and derivative prices.

In this work, we analyze inflation expectations using instruments from the Brazilian fixed income market, through a parametric affine term structure model of the Nelson–Siegel family. The article contributes by analyzing a market with unique characteristics, especially the widespread use of inflation indexed instruments. Through this model, we can decompose the breakeven inflation into expected inflation and risk premium using a no-arbitrage theoretical consistent formulation, avoiding ad-hoc specifications based on purely statistical models. To the best of our knowledge, this is the first article based on the no-arbitrage method proposed in Christensen, Diebold, and Rudebusch (2011) that accesses expected inflation in an emerging market.

The Brazilian fixed-income market has special features that justify this study. In particular, it is a market that was initially developed for protection against inflation in the hyperinflation period, and thus the indexed bond market was highly relevant from the beginning (Leal and Silva 2008), in contrast to developed markets where until recently indexed securities were less relevant. Another interesting point is that in the Brazilian bond market, the pricing starts in the futures markets, and then is transmitted to the fixed income securities, as discussed in Silva and Holland (2013). The use of no-arbitrage conditions in our parametric model is fundamental, since by imposing these conditions, we

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can extract the inflation expectations and the risk premium component in consistent form, without the ad-hoc or purely statistical specifications as in previous works on this topic.

The article is organized as follows: in the following section, we describe the Brazilian fixed income market and the financial instruments used in the empirical analysis. The Literature Review section describes the relevant literature on the extraction of inflation expectations and risk premiums present in fixed-income securities. In the section Nelson–Siegel Arbitrage-Free Model, we briefly present the model and how to decompose the interest rate differentials into expected inflation and risk premium. The estimation method and the estimated parameters are presented in the Empirical Approach section. The Results section presents the main results, the measures of model fit and forecast accuracy tests using future inflation, comparing the model's results with the forecasts of the Focus Bulletin survey conducted by the Brazilian Central Bank. The Conclusions section presents the closing remarks.

Brazilian Fixed Income Market

According to Leal and Silva (2008), the Brazilian fixed income market is one of the most liquid and developed among emerging markets. The auctions of LTN, LFT, and NTN-F, the notes and bills issued by the National Treasury, occur every week, and the inflation-indexed NTN-B bonds once a month. These articles are typically in American style (multiple price). The Treasury also holds repurchase auctions to provide liquidity and price references to the market, and also to adjust the debt profile to the requirements of the market and the Treasury. Bonds are traded in the secondary market in over-the-counter contracts and through the screen, by means of the Brazilian Mercantile and Futures Exchange —BM&F Bovespa and CETIP (Central de Custódia e de Liquidação Financeira de Títulos; Central Custody and Financial Settlement of Securities) systems. For a sense of the importance of this market, the level of Treasury bonds increased significantly during the economic stabilization process, from 6.47% of GDP in 1990, to 54.96% of GDP in 2005.

An essential feature in the Brazilian market is the importance of futures contracts in the determination of the entire interest rate market. The *Contrato Futuro de Taxa Média de Depósitos Interfinanceiros de um Dia—DI* (One Day Interbank Deposits Future) negotiated on the BM&F Bovespa is more actively traded in notional terms than fixed rate government bonds (Jeanneau et al. 2007), and is the main reference for the term structure of interest rates in Brazil. The contract is equivalent to a zero coupon bond financed at the interbank overnight rate and offers Treasury bond holders the opportunity to transfer their risk positions in a highly liquid environment at low transaction costs. The contract is cash settled, and is marked to market every business day.

As discussed in Silva and Holland (2013), the Brazilian fixed income market is an anomaly. Due to the high inflationary dynamics in Brazil in the period before 1995, futures markets evolved much faster than the spot bond markets, and the liquidity is much higher than the government bond market still today. This feature makes the DI yield curve the true reference curve for nominal yields in Brazil, in contrast to the general rule where the determination of the yield curve is based on the government bond market. As the DI future is a cash settled instrument, there is no need for physical delivery of the instrument. This enables high leverage in this market, leading to high liquidity in contrast to the existing market for government bonds and notes in Brazil. As the market is set via margin adjustments, is only required to deposit an initial guarantee set by the BM&F, and thus the necessary capital is smaller, compared to an equivalent operation in the Treasury market. In the Brazilian Treasury market it is necessary to establish a bank reserve for the traded bonds, which increases the operating costs. As pointed out by Silva and Holland (2013) this structure can induce migration of transactions involving interest rates to future markets.

Another key feature in this sense pointed by Silva and Holland (2013) is the fact that trading of LTNs (National Treasury Bills) and NTN-Fs (National Treasury Notes-F series) are locked to the negotiation of the DI futures. This mechanism serves to reduce exposure to interest rate risk through a hedge position for the purchase of notes with an interest rate lock mechanism in the DI market. Silva and Holland (2013) studied the determination of interest rates in Brazil and obtained strong indications

that these rates are determined first in the futures market and transmitted by no-arbitrage to the government bond market. Another important point discussed by these authors is that the DI contracts also represent the expectations for the Selic, the Central Bank reference of interest rates. The risk of NTN and LTN transactions is very close to zero, since the issuer is the federal government. Due to the margin requirement mechanisms of the BM&F, the default risk in DI futures contracts can be considered close to zero, and so the future interest rate contracts and the government bond market represent the same risk factor.

Due to these characteristics, we base our analysis on a nominal curve based on DI Futures contract and as the reference for indexed curves the yields from Brazilian Treasury NTN-B notes. The NTN-B is important because it is indexed to the same measure of inflation in the official inflation target, the Comprehensive National Consumer Price Index—IPCA, that serves as an anchor for the decisions of policymakers, besides being the Treasury bond with greatest liquidity in Brazil.

Most of the articles about Brazil use National Treasury Notes (NTNs) as reference for nominal yields. But, as the methodology of our work is fundamentally based on no-arbitrage conditions, which are essentially linked to the presence of liquidity in the market. As the DI futures market is more liquid, has lower transaction costs and allows greater leverage, we believe it is a more appropriate reference for an arbitrage free yield curve.

Our sample starts in 2006, due to the fact that the trading volume and stock of NTN-B (indexed bond) reached a significant level in this period, increasing its liquidity. The stock of NTN-Bs rose as a share of government debt securities from 6.7% in January 2004 to 64.1% in November 2013, becoming currently the main reference of debt securities in Brazil. The mean daily traded volume of NTN-B increased from US$ 6 million to more than US$ 3.5 billion between January 2004 and July 2013.

Literature Review

The interest rate differential between real and nominal bonds of the same maturity is known as breakeven inflation (BEI). The BEI rate for each maturity available can be defined by the difference between the yields of the nominal and indexed instruments. The BEI is an indicator of expected inflation, but, it is an imperfect indicator because it includes inflation risk premiums. The biggest problem is that the estimation of the risk premium in general depends on the specification of a parametric model, which defines the market risk premium for the uncertainty associated with future inflation. There are several studies using government bonds to extract expected inflation for different markets (e.g. Woodward (1990) and Deacon and Derry (1994) for the British market and Shen and Corning (2001) for the American market). Scholtes (2002) concluded that for the English case, the BEI rate has better empirical performance than inflation surveys. Christensen et al. (2004) found that indexed bonds indicate consistently higher expected inflation than survey expectations for the Canadian market, concluding that survey indicators are consistent estimates of inflation expectations and the breakeven rate (BEI) is a biased estimate due to the presence of risk premiums.

The presence of inflation risk premiums is a key motivation for the existence of indexed securities. Indexed bonds are important for diversification, risk sharing, and market completeness, since in general, the existing securities, nominal Treasury bonds and equities, are imperfectly correlated with (unexpected) inflation, being very poor hedges of inflation risk. Bekaert and Wang (2010) also found that the inflation premiums in the US Treasury inflation protected securities (TIPS) are sizable and time varying. Hördahl (2008) also observed that inflation risk premiums in the United States and the Euro area vary significantly over time, mainly in response to fluctuations in economic growth and inflation, and the results in Hördahl and Tristani (2010) indicate that cyclical dynamics of long-term inflation risk premia are mostly associated with changes in output gaps, while their high-frequency fluctuations seem to be aligned with variations in inflation. An important result of the inflation risk premium found in Söderlind (2011) is that breakeven deviations (the difference between breakeven

inflation and inflation expectations in survey data) can be explained by the presence of off-the-run liquidity premium and the inflation uncertainty present in data surveys.

The decomposition of the BEI rate into inflation risk premium and inflation expectations using a parametric model can be done through affine no-arbitrage models (e.g. Duffie and Kan (1996) and Filipovic (2009)). These models are commonly used to model term structure of interest rates. However, affine models can have poor fit to the data (e.g. Duffee (2002)), so alternative representations are frequently used in the fitting of yield curves. Based on the parameterization of Nelson and Siegel (1987), a dynamic formulation based on latent factors was proposed by Diebold and Li (2006). These models have a better in-sample fit and more accurate forecasts than affine models, and so, have been were extensively adopted in the modeling of yield curves, as discussed in Diebold and Rudebusch (2011). A affine version of the dynamic Nelson–Siegel model with the imposition of no-arbitrage conditions was proposed by Christensen, Diebold, and Rudebusch (2011), obtaining a model with both good empirical power and consistent with no-arbitrage conditions. This model can be used to decompose the BEI rate into implied inflation rate and risk premium, since it is based on an explicit market premium (price) of risk. Using this affine Nelson–Siegel formulation to analyze inflation expectations in the United States, Christensen, Diebold, and Rudebusch (2011) concluded that their estimates are close to the financial market surveys and risk premiums vary around zero. They also said that the use of this approach has the advantage of being easily estimated and updated by the Kalman filter, generating high frequency results which are desirable for policymakers and financial market professionals.

In this article, we use the parametric model with no-arbitrage conditions of Christensen, Diebold, and Rudebusch (2011) to decompose the BEI rate in expected inflation and risk premium in Brazil. For the Brazilian bond markets, parametric models are employed to extract the implicit inflation, but none of these imposes no-arbitrage restrictions. Thus, our work tries to get a more accurate measure of the expected inflation in financial markets based on no-arbitrage formulations of the yield curve, using a formulation based on the dynamic no-arbitrage extensions of the widely used Nelson–Siegel family of term structure models (Nelson and Siegel 1987; Christensen, Diebold, and Rudebusch 2011; Diebold and Rudebusch 2011).

There are several works analyzing inflation expectations in Brazil. In Vicente and Guillen (2013), the authors test whether breakeven inflation is a good measure of future inflation. With data from Brazilian government NTN and NTN-B notes, the authors perform forecast bias tests using univariate regression methods. The authors propose a regression of future inflation against expected inflation plus a constant. They find evidence that breakeven inflation is an unbiased estimator for short (3 and 6 months) and medium-term (12 and 18 months) horizons. This breakeven inflation has little explanatory power and for longer horizons (24 and 30 months), they find the counterintuitive result of a negative relationship between future inflation rates and the BEI.

Caldeira and Furlani (2014) estimated Svensson (1994) models with four latent factors for nominal and indexed notes in Brazil. With this model they found the BEI rate as the differential between model-derived nominal and real bond yields. They also adopted forecast bias tests similar to those used by Vicente and Guillen (2013), but also use a state-space formulation with a time-varying parameter representing the risk premium. Caldeira and Furlani (2014) find that the breakeven inflation is an unbiased estimator for inflation 3 months ahead and it provides information only for shorter horizons. They also compare the model predictions with VAR (vector autoregressive) models and market expectations in the Central Bank's Focus survey. The BEI rates are superior to VAR models proposed by the authors, but worse than the survey in terms of forecast accuracy. We can also cite the work of Val et al. (2011) which applies several different methods to estimate implicit inflation and compares it to the estimates of the BEI rate. The authors find a relatively low inflation risk premium in the Brazilian market, and that inflation surveys embody a greater risk than securities trading data.

Nelson–Siegel Arbitrage-Free Model

This section briefly describes the main aspects of the Nelson–Siegel arbitrage-free model used to analyze the daily data from DI Futures and NTN-B bonds. We follow the presentation of Christensen et al. (2010). The nominal yield can be decomposed by:

$$y_t^N(\boldsymbol{\tau}) = y_t^R(\boldsymbol{\tau}) + \boldsymbol{\pi}_t^e(\boldsymbol{\tau}) + \boldsymbol{\phi}_t(\boldsymbol{\tau}) \tag{1}$$

with $y_t^N(\boldsymbol{\tau})$ being the nominal bonds yield to maturity$\boldsymbol{\tau}$, $y_t^R(\boldsymbol{\tau})$, the real bond yield for the same maturity, $\boldsymbol{\pi}_t^e(\boldsymbol{\tau})$ the implicit inflation in the bond market in period t to period $t+\boldsymbol{\tau}$ and $\boldsymbol{\phi}_t(\boldsymbol{\tau})$ the risk premium to hold the nominal bond until $t+\boldsymbol{\tau}$. We can write the implicit inflation as:

$$\boldsymbol{\pi}_t^e(\boldsymbol{\tau}) = -\frac{1}{\boldsymbol{\tau}} \ln E_t^P\left[e^{-\int_t^{t+\boldsymbol{\tau}} (r_t^R - r_t^N)dt}\right] \tag{2}$$

and the risk premium as:

$$\boldsymbol{\phi}_t(\boldsymbol{\tau}) = -\frac{1}{\boldsymbol{\tau}} ln\left(1 + \frac{cov(\frac{M_{t+\boldsymbol{\tau}}^R}{M_t^R}, \frac{Q_t}{Q_{t+\boldsymbol{\tau}}})}{E_t^P\left[\frac{M_{t+\boldsymbol{\tau}}^R}{M_t^R}\right] \times E_t^P\left[\frac{Q_t}{Q_{t+\boldsymbol{\tau}}}\right]}\right) \tag{3}$$

Therefore, the risk premium of keeping a nominal title can be positive or negative. The risk premium will be positive only if the $cov(\frac{M_{t+\boldsymbol{\tau}}^R}{M_t^R}, \frac{Q_t}{Q_{t+\boldsymbol{\tau}}})<0$, i.e. an increase in the price level, which is a moment of loss of purchasing power, has a positive effect on the price of real bonds. We can decompose the BEI into implicit inflation and risk premium:

$$BEI(\boldsymbol{\tau})_t = y_t^N(\boldsymbol{\tau}) - y_t^R(\boldsymbol{\tau}) = \boldsymbol{\pi}_t^e(\boldsymbol{\tau}) + \boldsymbol{\phi}_t(\boldsymbol{\tau}) \tag{4}$$

Using the Nelson–Siegel structure, we can represent the observed yield curves by:

$$y(\boldsymbol{\tau})^N = L_t^N + S_t((1-e^{-\boldsymbol{\lambda}\boldsymbol{\tau}})/\boldsymbol{\lambda}\boldsymbol{\tau}) + C_t((1-e^{-\boldsymbol{\lambda}\boldsymbol{\tau}})/\boldsymbol{\lambda}\boldsymbol{\tau} - e^{-\boldsymbol{\lambda}\boldsymbol{\tau}}) + \boldsymbol{\varepsilon}_{N,\boldsymbol{\tau}} \tag{5}$$

$$y(\boldsymbol{\tau})^R = L_t^R + \boldsymbol{\alpha} S_t((1-e^{-\boldsymbol{\lambda}\boldsymbol{\tau}})/\boldsymbol{\lambda}\boldsymbol{\tau}) + \boldsymbol{\alpha} C_t((1-e^{-\boldsymbol{\lambda}\boldsymbol{\tau}})/\boldsymbol{\lambda}\boldsymbol{\tau} - e^{-\boldsymbol{\lambda}\boldsymbol{\tau}}) + \boldsymbol{\varepsilon}_{R,\boldsymbol{\tau}} \tag{6}$$

with L_t^N and L_t^R being the latent level factors of nominal and real bonds, respectively, S_t the slope factor, C_t the curvature factor and $\boldsymbol{\alpha}$ the adjustment term for real bonds. To facilitate the estimation, we will discretize this process as in Mouabbi (2013). Setting $X_t = (L_t^N, S_t, C_t, L_t^R)$, we can represent the transition equations as:

$$X_T = [I - e^{K^P(T-t)}]\boldsymbol{\theta}^P + e^{K^P(T-t)}X_t + \boldsymbol{\eta}_t \tag{7}$$

and thus the transition equation can be estimated in discrete time as a first order autoregressive process under a time lag $(T-t)$. This transformation allows us to estimate and recover the parameters of the measure P dynamics (for further details see Christensen et al. (2010)), and enables the estimation by quasi-maximum likelihood using the Kalman filter, e.g. Christensen et al. (2014).

Christensen et al. (2010) inflation show that, assuming a Gaussian likelihood, one can obtain an affine formulation for the yield differential:

Table 1. Descriptive statistics—NTN-B, DI-futures, and breakeven rates (BEI).

	Indexed yields (NTN-B)			
Maturity	**Mean**	**Standard deviation**	**Minimum**	**Maximum**
1 month	7.378	6.208	–1.775	29.812
3 months	7.021	4.609	–6.075	28.217
6 months	6.915	3.520	0.218	23.652
9 months	6.855	3.106	0.747	24.947
12 months	6.876	2.879	0.984	14.832
15 months	6.937	2.757	1.101	13.600
18 months	6.996	2.662	1.159	12.773
21 months	7.052	2.584	1.263	12.235
24 months	7.098	2.502	1.432	12.044
26 months	7.120	2.448	1.519	11.970
30 months	7.158	2.362	1.663	11.860
36 months	7.188	2.251	1.965	11.675
42 months	7.193	2.155	2.177	11.502
48 months	7.186	2.074	2.329	11.451
60 months	7.154	1.947	2.574	11.164
Source:ANBIMA				
Nominal Yields (DI-Futures)				
Month	12.355	3.549	6.926	19.823
Months	12.353	3.527	6.972	19.878
Months	12.398	3.460	6.915	19.734
Months	12.460	3.387	6.865	19.572
Months	12.541	3.304	6.864	19.424
Months	12.635	3.220	6.898	19.309
Months	12.723	3.135	6.985	19.690
Months	12.801	3.059	7.121	20.075
Months	12.870	2.994	7.306	20.412
Months	12.910	2.958	7.421	20.635
Months	12.972	2.899	7.627	21.098
Months	13.042	2.836	7.882	21.763
Months	13.099	2.788	8.105	22.419
Months	13.140	2.759	8.268	22.927
Months	13.189	2.729	8.512	23.568
Source:BM&F/Bovespa				
Breakeven Rate (BEI)				
Month	4.977	4.444	–13.186	17.405
Months	5.331	2.663	–8.527	25.235
Months	5.483	1.547	–4.632	10.331
Months	5.604	1.445	–5.887	17.880
Months	5.665	1.401	3.150	11.053
Months	5.698	1.397	3.286	10.747
Months	5.727	1.396	3.376	11.093
Months	5.748	1.400	3.394	11.425
Months	5.772	1.404	3.415	11.740
Months	5.790	1.413	3.378	11.945
Months	5.814	1.435	3.282	12.375
Months	5.854	1.465	3.394	13.019
Months	5.906	1.488	3.528	13.653
Months	5.954	1.508	3.628	14.138
Months	6.035	1.548	3.618	14.754

Notes: The *breakeven inflation* (BEI) is calculated as the spread between nominal and indexed yields of bonds of the same maturity.

$$E_t^P\left[e^{-\int_t^{t+\tau}(r_t^R-r_t^N)dt}\right] = exp[B^{\pi}(\tau)X_t + A^{\pi}(\tau)] \tag{8}$$

with $B^{\pi}(\tau)$ and $A^{\pi}(\tau)$ being the solutions of Riccatti differential equations, solved by the fourth order Runge–Kutta method. Using this representation we can calculate the implied inflation and risk premium. For more details, see Christensen et al. (2010) and Christensen, Diebold, and Rudebusch (2011).

Empirical Approach

The estimation is performed using daily data of the DI-Future and NTN-B yield curves, from 01/02/2006 to 10/01/2013, corresponding to 1,915 observations. The sources of the data are ANBIMA (Brazilian Association of Financial and Capital Market Entities) and BM&F Bovespa. We use the closing prices of future DI contracts negotiated in the BM&F Bovespa derivatives market, and the reference prices for the NTN-B (National Treasury Bills-Type B) bonds in Brazil provided by ANBIMA. To ensure consistency between the nominal and indexed yields, we use the reference curve used for the marking to market of NTN-B bonds provided by BM&F Bovespa, constructed using ANBIMA data on yields of NTN-B bonds. The data are publicly available from the BM&F Bovespa ftp server, at ftp://ftp.bmf.com.br.

The descriptive statistics of the indexed, nominal and BEI rates are presented in Table 1, for 15 maturities from one to 60 months. The NTN-B bond pays a coupon every 6 months and the first interest coupon paid includes the full rate for 6 months, regardless of the purchase settlement date. The NTN-B yields are transformed in zero coupon equivalent yields using a bootstrapping method. All yields are quoted on an annual basis. We used fixed maturities, constructed using a cubic spline interpolation based on the traded maturities on each day.

To find what best specification for the joint model for real and nominal bonds, we need to determine the number of latent factors in the model. In general, models of the term structure of interest rates use three factors: the level, slope and curvature. We follow Christensen et al. (2010) and use principal component analysis to determinate the number of factors, analyzing maturities of 1, 3, 6, 12, 15, 18, 21, 24, 30, 36, 42, 48, and 60 months for the nominal and indexed bonds. Four factors explain more than 99.2% of the curves. According to this result, we propose a model with: two level factors, one for each type of security, and common slope and curvature factors.[1]

Estimation

We estimate parameters and latent factors using the quasi-maximum likelihood methodology based on the Kalman filter, following Christensen et al. (2014). The measurement Equations (5) and (6) can be put in a state space representation as:

$$y_t = BX_t + \varepsilon_t \tag{9}$$

where X_t represents the adjusted Nelson–Siegel factors and ε_t represents the measurement equations' errors, which are assumed independent and identically distributed (IID) for each maturity. The state equation of the latent factors is given by Equation (7), and can be written as:

$$X_t = \Theta + \Phi X_{t-1} + \eta_t \tag{10}$$

and we assume an error structure given by:

$$\begin{pmatrix} \boldsymbol{\eta}_t \\ \boldsymbol{\varepsilon}_t \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} R & 0 \\ 0 & Q \end{pmatrix} \right) \tag{11}$$

where the state variance matrix $R = diag(\sigma_i^2)$ (4×4) and measurement equations have the same variance with $Q = diag(\boldsymbol{\sigma}^2)$. Furthermore, the optimality of the Kalman filter needs the errors to be orthogonal to the state variable: $E(X_t \boldsymbol{\eta}_t) = 0$ and $E(X_t \boldsymbol{\varepsilon}_t) = 0$.

The model must be both flexible and parsimonious, and it is essential that it be correctly specified. Therefore, we use several statistical tests and information criteria to choose the best specification. The results of these analyses are presented in Table 2. All information criteria—Akaike (AIC), Bayesian (BIC) and Hannan–Quinn indicate that the best specification is the diagonal specification for the latent factor structure.

Table 3 shows the estimated values for the chosen model. In this table, we present the estimates for the parameter matrices $\boldsymbol{\phi}$, $\boldsymbol{\theta}$, Σ in addition to the decay factor ($\boldsymbol{\lambda}$) and the adjustment parameter $\boldsymbol{\alpha}$ for the indexed yields. We estimate a decay ($\boldsymbol{\lambda}$) of 0.193, near the value of other works estimating dynamic Nelson–Siegel models for Brazilian yield curves. The parameter $\boldsymbol{\alpha}$ was estimated close to 0.97, similar to Christensen et al. (2010) who find a value of 0.92.

Results

In this section we present the estimated values of the latent factors and the fit measures of the model. We also show the estimated values for the implicit inflation and risk premiums for the Brazilian fixed-income market. Finally, we test if implicit inflation is a good predictor of future inflation.

Table 2. Model specification analysis.

	Tests				
	logL	Parameters	AIC	BIC	Hannan–Quinn
Unrestricted	–32612	27	34.106	34.184	34.135
1 Restriction	–32853	26	34.339	34.414	34.367
2 Restriction	–32836	25	34.320	34.392	34.346
3 Restriction	–32546	24	34.016	34.085	34.041
4 Restriction	–32569	23	34.038	34.105	34.063
5 Restriction	–32393	22	33.854	33.917	33.877
6 Restriction	–32339	21	33.796	33.857	33.818
7 Restriction	–32338	20	33.794	33.852	33.815
8 Restriction	–32339	19	33.794	33.849	33.815
9 Restriction	–32159	18	33.605	33.657	33.624
10 Restriction	–32161	17	33.606	33.655	33.624
11 Restriction	–31986	16	33.422	33.468	33.439
Diagonal	–31972	15	**33.407**	**33.450**	**33.423**

Notes: We imposed the least significant parameter was equal to zero in each constraint. Model with 1 constraint: $\phi_{1,4} = 0$. Model with 2 constraints: $\phi_{1,4} = \phi_{4,1} = 0$. Model with 3 constraints: $\phi_{1,4} = \phi_{4,1} = \phi_{4,3} = 0$. Model with 4 constraints: $\phi_{1,4} = \phi_{4,1} = \phi_{4,3} = \phi_{3,4} = 0$. Model with 5 constraints: $\phi_{1,4} = \phi_{4,1} = \phi_{4,3} = \phi_{3,4} = \phi_{2,4} = 0$. Model with 6 constraints: $\phi_{1,4} = \phi_{4,1} = \phi_{4,3} = \phi_{3,4} = \phi_{2,4} = \phi_{3,1} = 0$. Model with 7 constraints: $\phi_{1,4} = \phi_{4,1} = \phi_{4,3} = \phi_{3,4} = \phi_{2,4} = \phi_{3,1} = \phi_{4,2} = 0$. Model with 8 constraints: $\phi_{1,4} = \phi_{4,1} = \phi_{4,3} = \phi_{3,4} = \phi_{2,4} = \phi_{3,1} = \phi_{4,2} = \phi_{1,3} = 0$. Model with 9 constraints: $\phi_{1,4} = \phi_{4,1} = \phi_{4,3} = \phi_{3,4} = \phi_{2,4} = \phi_{3,1} = \phi_{4,2} = \phi_{1,3} = \phi_{2,1} = 0$. Model with 10 constraints: $\phi_{1,4} = \phi_{4,1} = \phi_{4,3} = \phi_{3,4} = \phi_{2,4} = \phi_{3,1} = \phi_{4,2} = \phi_{1,3} = \phi_{2,1} = \phi_{1,2} = 0$. Model with 11 constraints: $\phi_{1,4} = \phi_{4,1} = \phi_{4,3} = \phi_{3,4} = \phi_{2,4} = \phi_{3,1} = \phi_{4,2} = \phi_{1,3} = \phi_{2,1} = \phi_{1,2} = \phi_{3,2} = 0$.

Table 3. Estimated parameters—diagonal model.

	$\phi, 1$	$\phi, 2$	$\phi, 3$	$\phi, 4$	θ	α	λ
$\phi 1$,	0.995				–0.016	0.972	0.193
	(0.001)				(0.009)	(0.000)	(0.000)
$\phi 2$,		0.878			0.264		
		(0.002)			(0.007)		
$\phi 3$,			0.972		0.060		
			(0.003)		(0.017)		
$\phi 4$,				1.000	–0.007		
				(0.000)	(0.000)		

Notes: The table presents the estimated parameters for the diagonal specification chosen by the information criteria analysis.

Latent Factors and Fitting of Observed Yields

Figures 1 and 2 show the level, slope and curvature factor estimates for real and nominal yields for the period analyzed. The correlation between these two level factors is estimated as 91.71%, and we note that both the real and the nominal level factors show an increase in 2008, the period of the Lehmann Brothers crisis. The curvature and slope factors are much less volatile than the level, in line with the literature on term structure latent factors (e.g. Diebold and Li (2006)).

Table 4 shows measures of fit of the model for different maturities in terms of mean error and the root mean squared error. These measurements show that the model generally provides good fit, especially for indexed yields, as can be seen by the smaller root mean squared errors for these series.

Implicit Inflation and Risk Premiums

From Equation (8), we can calculate the implicit inflation derived from our model. As discussed in Section 4, the BEI rate can be decomposed into implicit inflation and risk premium. Using Equation (4) we can calculate the risk premium for each maturity. Using the same approach of Christensen et al.

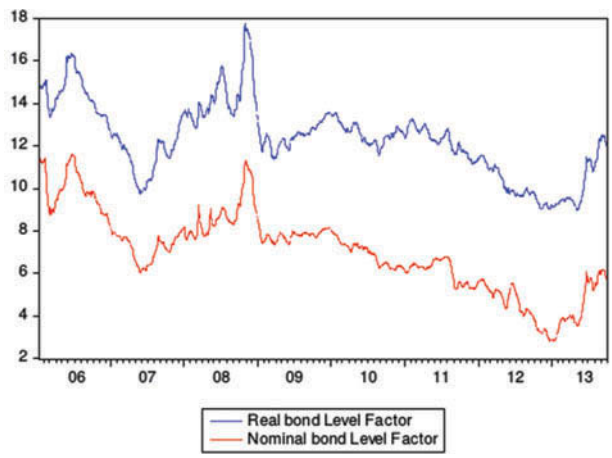

Figure 1. Estimated level factors.

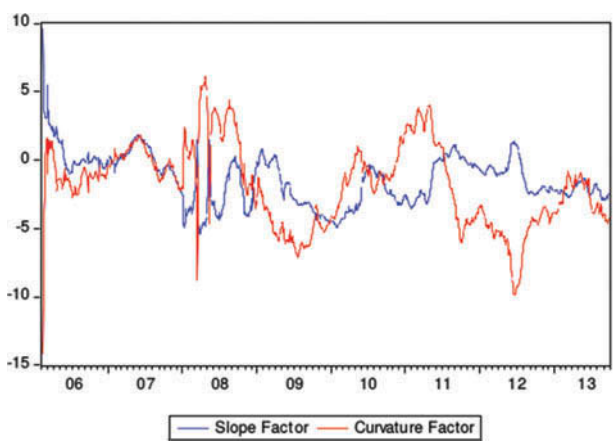

Figure 2. Estimated slope and curvature factors.

Table 4. Fit measures—observed yields.

	Mean error	Root mean squared error
Indexed		
Maturity 1 month	–0.095	2.798
Maturity 3 months	0.054	0.571
Maturity 6 months	–0.015	0.195
Maturity 9 months	–0.007	0.113
Maturity 12 months	0.001	0.086
Maturity 15 months	–0.001	0.077
Maturity 18 months	–0.000	0.074
Maturity 21 months	–0.002	0.077
Maturity 24 months	–0.007	0.079
Maturity 26 months	–0.009	0.081
Maturity 30 months	–0.007	0.083
Maturity 36 months	0.010	0.085
Maturity 42 months	0.036	0.092
Maturity 48 months	0.063	0.100
Maturity 60 months	0.128	0.139
Nominal		
Maturity 1 month	0.042	0.876
Maturity 3 months	0.026	0.405
Maturity 6 months	0.028	0.224
Maturity 9 months	0.046	0.199
Maturity 12 months	0.047	0.203
Maturity 15 months	0.028	0.209
Maturity 18 months	0.009	0.210
Maturity 21 months	–0.007	0.204
Maturity 24 months	–0.023	0.197
Maturity 26 months	–0.032	0.194
Maturity 30 months	–0.040	0.193
Maturity 36 months	–0.041	0.196
Maturity 42 months	–0.049	0.203
Maturity 48 months	–0.053	0.211
Maturity 60 months	–0.049	0.235

Notes: The table shows the mean error and root mean squared error between the model fitted and observed yields for the nominal and indexed instruments.

(2010), we compared the breakeven inflation and the implicit inflation extracted from the model with the market survey forecasts from Focus Bulletin and realized inflation in the analyzed period.

The Focus Bulletin is a survey released every Monday by the Central Bank of Brazil. The report, a daily survey of forecasts of about 120 banks, funds and other institutions summarizes the market expectations for the Brazilian economy. The Market Expectations System, which collects information reported in the Focus Bulletin, consists of a web interface where the agents report their expectations for the macroeconomic variables of interest. The system considers the last information reported in the past 30 days by the market players. If an agent does not revise his or her expectations in the last thirty days, this information is not considered in the calculation of statistics. Another important incentive in this system is the monthly release of the institutions with the best prediction results, indicating the top five predictors for each macroeconomic variable monitored in the system. In the present work, we use as the benchmark for the market expectation of future inflation rates the average inflation forecasts published in the Focus Bulletin.

As can be seen in Figures 3 and 4, the implicit inflation extracted from the model seems to be more accurate than the Focus forecasts for both the horizons of 6 and 12 months. Both the mean error and the mean squared error are smaller than those from the Focus forecasts, according to the results in Table 5. When comparing the BEI rates with the implicit inflation, that the implicit inflation also seems to be a superior predictor.

Figure 5 presents the model risk premium for maturities of 6, 12, 24, and 60 months. It can be noted that the risk premiums are higher for shorter horizons. The dynamics of risk premiums seem to be the same between the yields of up to 24 months, with only a difference of level. Another interesting result is that risk premiums seem to be time varying in both time and in maturities.

This large variation in risk premiums is similar to the results obtained in the works of Hördahl (2008) and Söderlind (2011). We can see an amplification of volatility and risk premiums in 2008, associated with the transmission of the global financial crisis. We also observe a remarkable increase in the volatility of risk premiums for longer maturities in 2010 and 2011. These effects can be associated

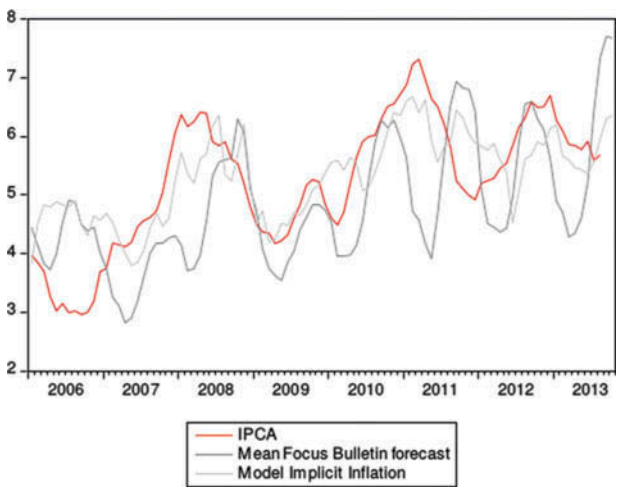

Figure 3. Focus bulletin, model expected inflation and IPCA performed 6 months ahead.

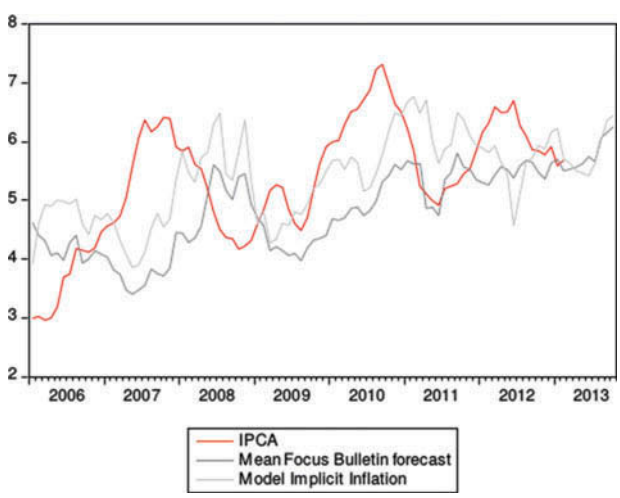

Figure 4. Focus bulletin, model expected inflation and IPCA performed 12 months ahead.

Table 5. Fit measures—observed inflation.

	6 months		12 months	
	ME	RMSE	ME	RMSE
Implicit Inf.	–0.049	0.592	0.006	1.127
BEIR	0.043	0.758	0.195	1.247
FOCUS[2]	0.454	1.471	0.619	1.456

Notes: The table shows the mean error and root mean squared error between the observed inflation and model implicit inflation, breakeven inflation rate and average Focus survey inflation forecasts, 6 and 12 months ahead.

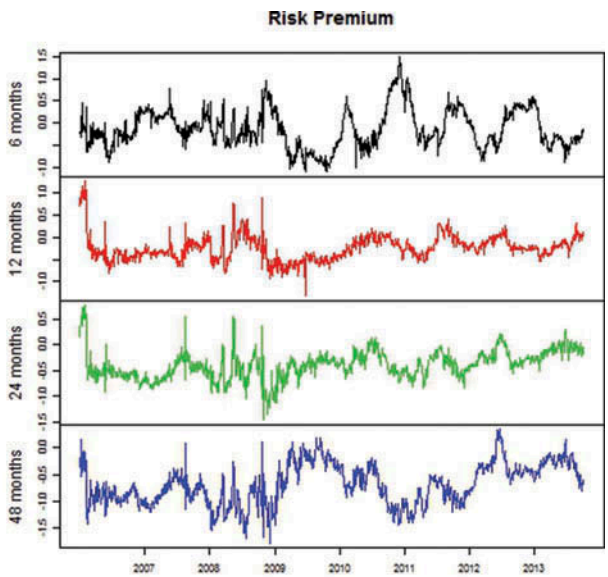

Figure 5. Risk premiums for maturities of 6, 12, 24, and 60 months.

with change in monetary policy started in 2010, with the use of several economic policy actions to reduce the effects of the economic slowdown observed in other countries, especially the risk of sovereign debt in the Euro zone and the rating downgrade of the US public debt in 2011. An explicit policy of reduction in interest rates set by the Brazilian government began in August 2011, accompanied by the elimination of macroprudential policies adopted earlier, such as credit restrictions and higher taxes on financial transactions.

The results in 2011 also indicate that increased volatility risk premium can be associated with inflationary pressures observed in this period, with inflation close to the target interval ceiling. In this way, we can also note that variations in the level and volatility of risk premiums might be associated with the weakening of the inflation targeting regime by the monetary authorities. Accumulated inflation rates in 2009 and 2010 were respectively 4.31% and 5.91%, versus a central target of 4.5%. The uncertainty associated with accelerating inflation seems to have a direct impact on the level of risk premiums of curves with maturities of 12, 24, and 48 months. This is most evident in 2011, where the accumulated variation of the IPCA in 2011 reached 6.50%. In 2012 and 2013, inflation was 5.84% and 5.91%, also close to the upper limit of the target band (6.5%). The reduction in interest rates starting in 2011, accompanied by elimination of macroprudential measures for credit

control and weaker commitment to the inflation target regime, seems to have had a direct impact on risk premiums in the period.

Implicit Inflation as a Predictor of Future Inflation

Vicente and Guillen (2013) and Caldeira and Furlani (2014) use Mincer–Zarnowitz regressions (Mincer and Zarnowitz 1969) to test the predictive power of implicit inflation and breakeven rates on future inflation. Defining $h(1)_t$ as the continuously compounded annualized inflation, we can define the cumulative inflation between period t and $t+\boldsymbol{\tau}$ as $h(\boldsymbol{\tau})_t = \frac{1}{\boldsymbol{\tau}} \sum_{j=t}^{j=t+\boldsymbol{\tau}} h_j(1)$. We can test whether the implied and breakeven inflation have predictive power on inflation using a regression with the following functional form:

$$h(\boldsymbol{\tau})_t = c_0 + c_1 \boldsymbol{\pi}_t^e(\boldsymbol{\tau}) + \boldsymbol{\varepsilon}_{t+\boldsymbol{\tau}} \tag{12}$$

where $\boldsymbol{\pi}_t^e(\boldsymbol{\tau})$ denote the predictors for future inflation, in this case the breakeven and implicit inflation rates extracted by the model.

Equation (12) implies that if the term c_1 is significant, then the inflation predictor used provides some information about future inflation, and if $c_0 = 0$ and $c_1 = 1$ the inflation predictor is an unbiased estimator of future inflation. Following Vicente and Guillen (2013) we first estimate Equation (12) by ordinary least squares (OLS). Later, to avoid possible endogeneity problems, we use estimates based on two-stage least squares (TSLS) using the first lag of the predictor (implicit inflation or breakeven rate) as an instrument, and generalized method of moments (GMM) estimations with the first three lags of the predictor as instruments.

Vicente and Guillen (2013) proposed estimation by TSLS and GMM to correct for a possible endogeneity problem, since future inflation should be influenced by inflation expectations. They also pointed to possible measurement errors in inflation expectations. Using these methods, we test the significance of c_1 and if $c_0 = 0$ and $c_1 = 1$ through Wald joint tests. Another point worth noting is that Vicente and Guillen (2013) said their estimates were correct only under the assumption that risk premiums are constant over time, which does not seem to be true for the Brazilian case.

Tables 6, 7 and 8 present the test results using the predictive regressions using model implicit inflation as predictor, and Tables 9, 10 and 11 using the breakeven rates as predictors of future inflation. It can be seen that, in general, the results are robust to estimation method, except for the BEI rates, where the results depend on which estimation method is used.

The implied inflation seems to be an unbiased estimator for up to 9 months, showing better results than those found in the works of Vicente and Guillen (2013) and Caldeira and Furlani (2014). The results also corroborate that for the medium term (24–48 months) the interest rate yields do not provide consistent information on future inflation. The results for implicit inflation for 36 months indicate that it is an unbiased estimator of inflation. For the long term (60 months), implicit inflation seems to provide some information, explaining more than 20% of the observed variation in inflation. The results of the Wald tests for the BEI rates in general indicate the rejection of the null hypothesis of unbiased prediction, and the estimated R^2 values are lower than those obtained using implicit inflation.

We conducted some further robustness analyses, checking if the results change with the inclusion of controls for the financial crisis observed in 2008. For this, we re-estimated the previous analyses including dummy variables for 2008 in the regressions. The results, available by request, are very similar to those obtained without controls, indicating that the overall results are robust to events observed in 2008.

Table 6. OLS estimation—observed inflation vs. implicit inflation.

	Maturity							
	3	6	9	12	24	36	48	60
c_0	1.154	1.103	0.827	0.426	0.101	0.652	–0.525	0.404
	[0.000]	[0.0000]	[0.0000]	[0.004]	[0.411]	[0.000]	[0.001]	[0.002]
c_1	–1.021	–0.594	0.882	3.077	5.086	2.037	8.715	3.751
	[0.077]	[0.327]	[0.227]	[0.000]	[0.000]	[0.002]	[0.000]	[0.000]
R^2	0.567	0.514	0.301	0.092	0.009	0.353	0.243	0.238
F-test	0.017	0.550	0.423	0.001	0.000	0.005	0.000	0.000

Notes: The table shows the regression results comparing the observed inflation to the implicit inflation, estimated ordinary least squares using Newey–West's adjusted robust standard errors. The values below the coefficients are the *p*-value of coefficients.

Table 7. TSLS estimation—observed inflation vs. implicit inflation.

	Maturity							
	3	6	9	12	24	36	48	60
c_0	1.351	1.178	0.838	0.370	0.158	0.839	–0.720	0.498
	[0.000]	[0.000]	[0.000]	[0.030]	[0.269]	[0.000]	[0.000]	[0.002]
c_1	–2.096	–0.994	0.840	3.403	4.788	0.994	9.824	3.209
	[0.002]	[0.160]	[0.321]	[0.000]	[0.000]	[0.191]	[0.000]	[0.001]
R^2	0.568	0.507	0.279	0.069	–0.003	0.359	0.308	0.234
F-test	0.001	0.334	0.565	0.002	0.000	0.244	0.000	0.000

Notes: The table shows the regression results comparing the observed inflation to the implicit inflation, estimated by two-stage least squares. Instruments—first lag of implied inflation. The values below the coefficients are the *p*-value of coefficients.

Table 8. GMM estimation—observed inflation vs. implicit inflation.

	Maturity							
	3	6	9	12	24	36	48	60
c_0	1.196	1.053	0.762	0.270	–0.016	0.697	–0.700	0.405
	[0.000]	[0.000]	[0.000]	[0.168]	[0.925]	[0.000]	[0.000]	[0.022]
c_1	–1.388	–0.292	1.266	3.996	5.862	1.596	9.679	3.815
	[0.172]	[0.787]	[0.248]	[0.001]	[0.000]	[0.089]	[0.000]	[0.000]
R^2	0.557	0.499	0.267	0.053	–0.014	0.343	0.343	0.228
F-test	0.020	0.963	0.450	0.001	0.000	0.194	0.000	0.001
Sargan test	0.698	0.909	0.988	0.403	0.868	0.644	0.778	0.665

Notes: The table shows the regression results comparing the observed inflation to the implicit inflation, estimated by the generalized method of moments. Instruments—first three lags of implied inflation. The values below the coefficients are the *p*-value of coefficients. The table reports the *p*-values values for the Sargan Test.

Conclusions

The aim of this study was to use the nominal and real interest rate markets in Brazil to extract expectations about future inflation, using the DI-Futures contract and NTN-B indexed bonds. Based on

Table 9. OLS estimation—observed inflation vs. breakeven inflation.

	Maturity							
	3	6	9	12	24	36	48	60
c_0	0.528	0.754	0.712	0.382	0.060	0.764	–0.439	0.406
	[0.000]	[0.000]	[0.000]	[0.004]	[0.625]	[0.000]	[0.008]	[0.007]
c_1	2.414	1.323	1.604	3.389	5.332	1.773	8.018	4.034
	[0.000]	[0.009]	[0.009]	[0.000]	[0.000]	[0.007]	[0.000]	[0.000]
R^2	0.311	0.416	0.305	0.094	0.003	0.389	0.138	0.186
F-test	0.000	0.033	0.026	0.000	0.000	0.000	0.000	0.000

Notes: The table shows the regression results comparing the observed inflation to the breakeven inflation, estimated by ordinary least squares using Newey–West's adjusted robust standard errors. The values below the coefficients are the *p*-value of coefficients.

Table 10. TSLS estimation—observed inflation vs. breakeven inflation.

	Maturity							
	3	6	9	12	24	36	48	60
c_0	0.725	0.834	0.712	0.313	0.089	0.922	–0.610	0.511
	[0.000]	[0.000]	[0.000]	[0.029]	[0.504]	[0.000]	[0.001]	[0.006]
c_1	1.351	0.906	1.623	3.773	5.199	0.961	8.902	3.510
	[0.023]	[0.128]	[0.019]	[0.000]	[0.000]	[0.184]	[0.000]	[0.000]
R^2	0.293	0.403	0.303	0.092	0.001	0.380	0.152	0.174
F-test	0.024	0.303	0.038	0.000	0.000	0.000	0.000	0.000

Notes: The table shows the regression results comparing the observed inflation to the breakeven inflation, estimated by two-stage least squares. Instruments—first lag of breakeven inflation. The values below the coefficients are the *p*-value of coefficients.

Table 11. GMM estimation—observed inflation vs. breakeven inflation.

	Maturity							
	3	6	9	12	24	36	48	60
c0	0.561	0.631	0.660	0.344	–0.021	0.839	–0.470	0.425
	[0.000]	[0.000]	[0.000]	[0.064]	[0.900]	[0.000]	[0.005]	[0.052]
c1	2.160	2.055	1.933	3.672	5.885	1.228	8.190	3.988
	[0.000]	[0.012]	[0.037]	[0.000]	[0.000]	[0.198]	[0.000]	[0.001]
$\hat{R}2$	0.310	0.386	0.306	0.092	–0.016	0.372	0.199	0.180
F-test	0.000	0.042	0.094	0.001	0.000	0.003	0.000	0.000
Sargan test	0.974	0.621	0.976	0.405	0.881	0.867	0.517	0.295

Notes: The table shows the regression results comparing the observed inflation to the breakeven inflation, estimated by the generalized method of moments. Instruments—first three lags of breakeven inflation. The values below the coefficients are the *p*-value of coefficients. The table report the *p*-values values for the Sargan Test.

the work of Christensen et al. (2010), we estimated an arbitrage free model for the joint dynamics of nominal and indexed curves, using a flexible structure with four factors: two different level factors and common slope and curvature factors.

Our article is original because it is the first work that, using this flexible parametric framework, imposes no arbitrage conditions for extracting inflation risk premiums in an emerging market. The results presented here for both the implied inflation rates and the BEI rate show better forecasts for future inflation 6 and 12 months ahead, compared to average market forecasts presented in the Focus Bulletin. When comparing the forecast accuracy of the predictions of the BEI and the implicit inflation, the measures of expect inflation derived from the no-arbitrage Nelson–Siegel model perform better for the period under analysis.

The results show that imposing no-arbitrage restrictions increases the predictive power of the model. The results also indicate that the Central Bank and economic agents benefit from the use of a model with the imposition of no-arbitrage conditions in the modeling of future inflation and the extraction of the inflation risk premium. The parametric model used performs better in predicting future inflation, which is especially important for inflation hedging and the correct pricing of nominal bonds not indexed to inflation.

Furthermore, the results indicate the presence of time-varying risk premium, invalidating the direct use of BEI rates in forecasting future inflation. This test also indicates that for shorter horizons the forecasts of the implicit inflation are unbiased estimators of future inflation, a result valid for horizons up to 9 months. As for medium-term and long-term horizons, the results are inconclusive, as already shown in other studies applied to the Brazilian market.

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